Individual Reflection

This project was completed individually, so I was responsible for both the Data Engineer and ML Engineer roles. The data was profiled and cleaned, then I created a time-based train/test split, built the sparse customer-product matrix, and documented the effect of each cleaning step. On the ML side, Truncated SVD and PCA were compared, tested, and evaluated against the best-seller baseline, and customer segments were created for the marketing use case.

The biggest learning point was that the quality of the final model is heavily dependent on the decisions made before the model is trained. For example, identifying non-product codes such as carriage and packing charges mattered because otherwise these could become products that the model recommends. I also saw why sparse data changes the choice of dimensionality-reduction method. Truncated SVD proved to be a better fit for the customer-product matrix because it works directly with sparse data without centering it.

If I changed my approach to this project, I would evaluate Hit Rate@10 across the different values of k, rather than choosing k mainly based on reconstruction error, runtime, and model size. This would allow recommendation performance to have a more direct role in choosing the final number of components. I would also test other recommendation models in addition to PCA and SVD to see whether they could improve Hit Rate@10 while still keeping the system fast enough for nightly use. Overall, this project helped me understand PCA and SVD as practical tools rather than only mathematical techniques.